

Review of the manuscript titled

Exhaustivity and contrastivity in Hungarian focus constructions: A visual-world study

The paper presents the results of an experiment that combines self-paced reading of stimulus sentences with eye-tracking in a two-image visual world task involving sentence-picture matching. The key aim of the study is to test whether the exhaustive interpretation of syntactically marked, unmodified pre-verbal focus (PVF) in Hungarian is semantic or pragmatic in nature. To adjudicate between these two existing approaches in the literature, the experiment compares stimulus sentences containing an unmodified PVF with sentences involving a PVF associated with the exclusive focus particle ‘only’ and with sentences in which PVF is preceded by the modifier ‘among others’.

The authors show that the sentence-picture matching task reveals a high rate of exhaustive interpretations of unmodified PVF, which is near-identical to the ‘only’-PVF condition, but differs from the ‘among others’-PVF condition. RT, reading time and eye-tracking data point to extra processing costs and more hesitation in the unmodified PVF condition compared to the ‘only’-PVF condition. In the ‘among others’ condition, in more than half of the cases participants selected the “non-exhaustive” image, while in more than one third of the cases the “exhaustive” image was chosen. The authors interpret these main results as supporting an account according to which the exhaustivity of unmodified PVF is due to pragmatic inference, which nevertheless leads to a robust bias toward the exhaustive interpretation.

The topic has broad linguistic interest and the experiment yields a rich set of potentially relevant, novel experimental data. The authors engage well with key literature, though this review is somewhat Hungarian-centric.

The manuscript is logically structured, very well written, and easy to follow.

Despite its merits, however, several crucial methodological weaknesses make it hard to recommend the paper for publication in JoL. I will detail these below.

1. Comparability of the conditions

The contrastive focus condition differs from the other two in non-trivial ways. Crucially, it has no “As for...” introduction of a topic. In the other two conditions a topic is introduced by “As for...”, and this topic delimits the range of alternatives that the focus of the sentence invokes.

This is a problematic confound for at least two reasons. First, topics guide attention and activate semantic fields that are not activated without them. This effect enters the processing of sentences in uncontrolled ways in the experiment, which is problematic in the case of the reading times and eye tracking measures.

Second, in the specific case in (6) and in examples like it, the introduction of the topic is likely to influence the interpretation of the comment that follows it. Out of context, in

the absence of a verbal prefix the comment may be interpreted either as narrow focus (on the pre-verbal object) or as broad, VP-focus. The topic 'as for drinks' raises the likelihood of a narrow focus reading in the case of (6a) and (6b). As there is no such topic in (6c), a broad focus interpretation is readily available there. This confounds any comparisons that involve this condition, including both online and offline measures.

If I understood correctly, half of the items had a verbal prefix IN EACH CONDITION. If so, that half of the items in the contrastive condition got disambiguated syntactically by the post-verbal placement of this prefix. That may (at least to some degree) counteract the absence of the topic in the 'contrastive' condition for the purposes of the offline interpretation task.

But even so, the effect of the absence vs the presence of the topic is quite unlikely to be neutralized in online processing measures.

In short, the confounding difference in the information structural makeup of the first two conditions vs the third condition makes them incomparable.

2. Participant assignment

I'm not sure I understand the design of how participants were assigned to different "parts" of the experiment, but if I understand it correctly, the arrangement is quite unusual. It sounds as if the authors had run an experiment third the size of what they call their experiment, but this smaller experiment was conducted three times as many times. This is due to the author's decision to allow participants to take part in the smaller experiment (which the authors call a "part" of the experiment) two times or three times, each time getting one third of the full set of items (which are better called lists).

I am assuming that each of the three "parts"/lists contained the same number of items from each test condition and the same number of fillers in a Latin square design. At least, that's what the last sentence on page 7 suggests (though that is not completely clear either).

Allowing participants to complete varying numbers of lists invalidates the Latin square design for several reasons.

The primary goal of creating three distinct lists in a Latin square design is to prevent within-subject repetition of target items. If a participant completes two or three lists, s/he will encounter the exact same lexicalization multiple times across different conditions. This introduces repetition priming, carryover effects and hypothesis awareness, and it does so in an unbalanced way, since some participants took one list, some took two, and some took three. Choosing to participate more times (getting presented with more lists) also introduces self-selection bias.

Combining these data in a single standard LMM breaks statistical assumptions regarding equal exposure. Participants who do 108 trials (three lists) experience fatigue

and task adaptation that participants doing 36 trials (one list) don't. The residual variance for three-list participants will likely be much higher (or systematically drift over time) compared to one-list participants. But standard LMMs assume homoscedasticity of residuals across all subjects.

The Missing at Random (MAR) assumption required for unbiased parameter estimation in LMMs is also violated, since some data points are sampled at random from the population (those from one-list participants), but other data points (from two- or three-list participants) are sampled from a non-random subset of the population.

This design flaw essentially contaminates the authors' data, and this is especially relevant to reading times and eye-tracking, since second and third exposures completely alter natural reading and processing behavior.

The best the authors can do, I think, are one of two things:

- They can discard all second- and third-exposure data, and run the "one-third experiment" in a Latin square design with more participants (two times as many are needed as are left after discarding contaminated data).
- If the "one-third sized" experiment is too short (if one would need more data points per participant than that allows), then the authors may need to rerun the experiment so that all participants see all the material (all three lists).

3. The task

Participants were told that in each trial, after reading the stimulus sentence they should select the picture that best matches the sentence. As in the 'only'-PVF sentences only a single object is referred to, multi-object images do not "match" the "meaning of the sentence". The same holds of unmodified-PVF sentences. As opposed to these two, in contrastive PVF sentences reference is made to multiple objects (through the addition of 'among others').

Crucially, this alone may explain, on the one hand, the key differences between the last condition on the one hand, and the first two conditions on the other, as well as the key similarities between the first two conditions.

In particular, the first key result of the experiment was the high rate of single-object image selections not only in the 'only'-PVF condition, but also in the unmodified PVF condition, in clear contrast to the 'among others'-PVF condition. This is not surprising, since stimulus sentences in the first two conditions only refer to a single object, while in the contrastive condition the sentence refers to multiple objects (in example (6c), "This morning Esther drank coffee, among other things").

Further, while sentences in the contrastive condition made reference to multiple objects, they included explicit mention of just one object (coffee in the case of (6c))., The task instruction to choose the image that best matches the "meaning of the sentence" may be interpreted in at least two ways. First, the better matching image may be the one that contains the single object that is explicitly mentioned in the sentence.

Second, the better matching image may well be considered to be the one that contains multiple objects, as ‘among others’ sentences made reference to multiple (at least three) objects: the one that is explicitly mentioned and a plurality (so, at least two) of “others”. This may explain another key result, specifically, that in the ‘among others’ condition in more than half of the cases participants selected the multiple-object (“non-exhaustive”) image, but in more than one third of the cases they opted for the single-object (“exhaustive”) picture. That may have been the function of how the instruction to select the image that “best matches the meaning of the sentence” is interpreted.

In fact, this seems a more plausible explanation for the relevant result than what the authors propose. What they claim is that this pattern of image-selection in the ‘among others’ condition is due to the robustness of the exhaustive interpretation associated with the PVF position. That seems neither likely nor coherent: the exhaustive, single-object interpretation is simply incompatible with the meaning of the core proposition the sentence conveys. In the case of (6c), for instance, this core proposition entails that Esther drank several (kinds of) drinks (at least three, as noted above). The sentence, then, cannot at the same time be interpreted as entailing that she only drank one (kind of) drink, which would be the exhaustive interpretation of PVF.

In brief, the formulation of the task itself raises serious concerns regarding task validity.

Because the methodological concerns outlined above affect the core findings of the study, I have limited my review to these high-level issues for now. Addressing these fundamental points will likely require substantial changes to content, potentially leading to a very different paper.

My recommendation is a resubmission, and I look forward to evaluating a thoroughly revised manuscript that addresses these primary concerns.